Regime-Dependent Cross-Sectional Predictability in US Equities: A Comparison of LSTM and Classical Machine Learning

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***Abstract*—We evaluate whether Long Short-Term Memory (LSTM) networks outperform classical machine learning for forecasting daily United States equity returns, using a panel of 518 S&P 500 constituents spanning March 2005 to July 2026 sourced from Bloomberg. Under a strictly chronological protocol with an explicit shuffled-target leakage control, we find that no model, LSTM included, predicts next-day return *levels* with an accuracy exceeding a naive zero-return baseline. Reframing the task cross-sectionally, that is, predicting which stocks will out- or under-perform the daily universe mean, reveals statistically significant predictability that is concentrated in high-volatility regimes: a gradient-boosted tree attains a mean daily rank information coefficient (IC) of 0.044 ($t = 3.2$) on high-volatility days versus 0.005 ($t = 0.9$) on calm days, and the nonlinear model overtakes linear models specifically under stress. However, this predictability is a one-day phenomenon. A decile long-short portfolio earns a gross annualized Sharpe ratio of 2.09 in high-volatility periods but does not survive transaction costs at daily rebalancing (net Sharpe negative for every configuration), and the signal decays to an information coefficient of approximately zero at a five-day horizon, so slower rebalancing removes both the cost and the edge. We conclude that the detectable structure is a short-horizon, stress-concentrated effect that is real but not economically exploitable at the frequencies studied, and we argue that this negative-but-rigorous result is the appropriate scientific conclusion for the setting.**

***Index Terms*—LSTM, machine learning, equity return forecasting, cross-sectional momentum, market regimes, transaction costs, information coefficient**

I. Introduction

Deep learning has been widely proposed for financial time-series forecasting, and recurrent architectures such as the LSTM [1] are natural candidates given the sequential structure of price data. A recurring difficulty in this literature is that daily equity returns are close to a martingale difference sequence: their conditional mean is small relative to their volatility, and simple benchmarks are hard to beat out-of-sample [2], [6]. Reported successes frequently arise from subtle look-ahead leakage, survivorship bias, or evaluation that ignores transaction costs.

This paper asks a deliberately narrow question with a rigorous answer: on a broad, recent panel of US large-capitalization equities, does an LSTM provide any edge over classical machine learning for daily return prediction, and if any signal exists, is it economically meaningful? Our contribution is threefold. First, we establish that daily return-*level* prediction is a null result across eleven model families including an LSTM, validated by a shuffled-target control that confirms the absence of leakage. Second, we show that a cross-sectional reformulation uncovers genuine, statistically significant predictability that is concentrated in high-volatility regimes and is stronger for nonlinear models. Third, through a horizon-and-cost analysis we demonstrate that the effect is a one-day phenomenon that survives neither transaction costs at daily frequency nor the passage of time to a weekly horizon. The result is a complete characterization of where the signal lives and why it is not tradeable.

II. Related Work

Fischer and Krauss [3] apply LSTMs to S&P 500 constituents and report profitability before costs, while Krauss *et al.* [4] study deep networks and gradient boosting for statistical arbitrage. Gu, Kelly and Xiu [2] provide a comprehensive empirical asset-pricing benchmark and emphasize that tree ensembles and neural networks capture cross-sectional rather than time-series structure. The short-horizon reversal effect we ultimately isolate is consistent with the microstructure and liquidity-provision literature [5], in which apparent short-term predictability is compensation for bearing inventory risk and is eroded by trading frictions. Our methodological posture, in particular the emphasis on point-in-time data, chronological validation and explicit leakage tests, follows de Prado [6].

III. Data

The dataset comprises daily open, high, low, close and volume observations for the S&P 500 membership, exported from a Bloomberg Terminal via a custom staged export pipeline. After cleaning, the panel contains 2,448,678 stock-day observations across 518 tickers from March 2005 to July 2026. Prices are unadjusted terminal fields; corporate-action treatment is discussed in Section VI. A parallel context dataset

of twelve macro-financial series (equity indices, the VIX and MOVE volatility indices, the dollar index, the 2-, 10- and 30-year Treasury yields, and front-month crude oil, gold and copper) is used for regime definition and auxiliary features.

To keep development independent of terminal access, an open-source mirror was built from public sources with an identical output schema, and a parity check against the Bloomberg closes gave a median absolute difference of essentially zero across overlapping tickers, with the few larger discrepancies attributable to corporate-action conventions.

IV. Methodology

A. Features and Target

All predictive features are constructed from information available at the close of day t. They include multi-horizon log returns (1, 5, 21 days), 21-day realized volatility, an intraday high-low range, price-to-moving-average deviations, and a log-volume z-score. For the cross-sectional task we additionally compute the daily percentile rank of each momentum, volatility and volume feature across the universe. Macro-context features (volatility-index levels and changes, a Treasury term spread, and commodity momentum) are appended by trading date.

The level target is the next-day log return, $r_{i,t+1}$. The cross-sectional target is the daily demeaned return,

$$\tilde{r}_{i,t+1} = r_{i,t+1} - \frac{1}{N_t} \sum_{j=1}^{N_t} r_{j,t+1}, \tag{1}$$

which isolates relative performance and is dollar-neutral by construction.

B. Splits and Leakage Protocol

Data are partitioned strictly chronologically: training through 2021, validation in 2022, and testing from 2023 onward (466,426 test observations). Feature scalers and all hyperparameters are fit on training and validation data only. As a falsification test we retrain a representative model on *shuffled* training targets; a non-trivial out-of-sample IC in that setting would indicate leakage. The control returns IC $= 0.004$ ($t = 0.67$), statistically indistinguishable from zero, supporting the integrity of the pipeline.

C. Models

We compare eleven families: a zero-return and a momentum naive baseline; ordinary, ridge and elastic-net linear regression; a decision tree, random forest, XGBoost and LightGBM; a linear support-vector regressor; a degree-two polynomial ridge; and an LSTM. The reported LSTM is a compact single-layer network (hidden size 32, dropout 0.1) trained on 30-day sequences pooled across the panel with shared weights, optimized with Adam under early stopping on validation loss (patience two) and gradient clipping. A higher-capacity two-layer variant (hidden size 64, dropout 0.2, patience five) was also trained; it reached a lower validation loss but its test metrics were not retained, so no result is reported for it, and the level-prediction conclusion below holds for both configurations. Tree and boosting models use early stopping on the validation set; linear and kernel models are standardized within a pipeline.

TABLE I
Daily Level Prediction, Test Set (2023–2026)

Model	Dir. Acc.	IC	IC t-stat
SVR (linear)	0.490	0.0141	2.42
Ridge / Linear	0.491	0.0109	1.86
XGBoost	0.497	0.1413	1.83
LSTM	0.480	0.0042	0.76
Shuffled-target (ctrl)	0.498	0.0039	0.67
Naive zero	—	—	—
LightGBM	0.496	−0.0295	−0.69
Decision tree	0.497	−0.0838	−1.60

D. Evaluation

Beyond root-mean-square error and directional accuracy, our primary metric is the mean daily cross-sectional rank IC, the Spearman correlation between predicted and realized returns computed per day and averaged, with a t-statistic across days. For the economic assessment we form a daily decile long-short portfolio, taking equal-weighted long and short positions in the top and bottom prediction deciles. We report gross and net annualized Sharpe ratios, maximum drawdown, and average one-way turnover. Costs are charged at 10 basis points per side on the one-way turnover $\sum_i |w_{i,t} - w_{i,t-1}|$ at each rebalance.

V. Results

A. Daily Level Prediction is a Null Result

Table I reports test-set performance for the level target. No model materially beats the naive zero baseline on RMSE, and directional accuracy sits near 50% throughout. The strongest information coefficients belong to the linear and support-vector models (IC ≈ 0.011–0.014, $t \approx 1.9$–2.4), which are marginal, while the LSTM attains IC $= 0.004$ ($t = 0.76$), statistically identical to the shuffled-target control. Gradient boosting and the decision tree have negative point-estimate ICs. The headline conclusion is unambiguous: for next-day return levels, an LSTM offers no advantage, and the task is effectively unforecastable with these features.

B. Cross-Sectional Predictability Concentrates in High-Volatility Regimes

We define a high-volatility regime as any day on which the 63-day z-score of the VIX exceeds one; this labels 18.7% of days and uses only information through day t. Table II reports the cross-sectional results by regime. Two findings stand out. First, predictability is strongly regime-dependent: LightGBM achieves IC $= 0.044$ ($t = 3.2$) on high-volatility days against 0.005 ($t = 0.9$) on normal days, and the pooled significance masks this concentration. Second, the nonlinear model overtakes the linear model *only* under stress, indicating nonlinear cross-sectional structure that pays off when volatility is high.

TABLE II
CROSS-SECTIONAL DAILY RESULTS BY REGIME, TEST SET

Model	Regime	IC	IC t	$\text{Sharpe}_{\text{gr}}$	$\text{Sharpe}_{\text{net}}$
Ridge	high-vol	0.0320	1.87	0.96	−0.69
Ridge	normal	0.0111	2.04	0.28	−2.93
Ridge	all	0.0154	2.76	0.48	−1.99
LightGBM	high-vol	0.0436	3.17	2.09	−0.88
LightGBM	normal	0.0048	0.88	−0.24	−4.89
LightGBM	all	0.0127	2.44	0.44	−3.55

TABLE III
FIVE-DAY HORIZON, WEEKLY REBALANCE (NON-OVERLAPPING)

Model	Regime	IC	IC t	$\text{Sharpe}_{\text{net}}$
Ridge	high-vol	−0.0071	−0.19	−0.27
Ridge	normal	0.0027	0.20	−0.74
Ridge	all	0.0007	0.05	−0.64
LightGBM	all	degenerate (no splits)		

An auxiliary experiment shows that the macro-context features contribute little beyond the regime split itself: ridge attains a comparable high-volatility IC with and without context, so it is the conditioning on regime, not the macro features, that matters.

C. The Signal is Short-Horizon and Not Cost-Survivable

The gross portfolio results are economically striking: the LightGBM high-volatility sleeve earns a gross annualized Sharpe of 2.09 with a shallow drawdown, and Fig. 1 shows its cumulative long-short return accruing almost entirely within high-volatility windows. The effect nonetheless fails two independent tests of economic reality. First, transaction costs eliminate it: average one-way daily turnover is roughly 2.6, implying about 26 basis points of daily cost, which exceeds the edge and drives every net Sharpe below zero (Table II). Second, the signal does not persist. Retargeting to a five-day forward return and rebalancing weekly with non-overlapping observations collapses the high-volatility IC to -0.007 (Table III); the gradient-boosted model degenerates to near-constant predictions (best iteration 1, prediction standard deviation 1.4×10^{-4}), finding no exploitable structure at that horizon. Slower rebalancing therefore removes the cost burden and the signal simultaneously.

VI. DISCUSSION AND LIMITATIONS

The pattern we observe, short-horizon relative predictability that intensifies in volatile markets and is competed away by trading costs, is consistent with short-term reversal as compensation for liquidity provision [5]. Its concentration in high-volatility regimes is intuitive: dispersion and the reward for absorbing order-flow imbalance both rise with volatility.

Several limitations bound our claims. The universe is current-membership S&P 500, which introduces survivorship bias; because such bias typically *flatters* performance, the true exploitable edge is if anything smaller than reported,

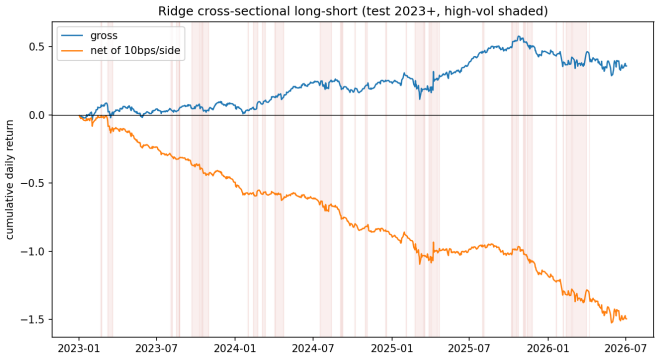


Fig. 1. Cumulative daily cross-sectional long-short return for the ridge model on the test set. Shaded bands mark high-volatility regimes. Gains accrue almost exclusively during stress periods; the net-of-cost line is materially below the gross line.

which strengthens rather than weakens the negative economic conclusion. A point-in-time membership reconstruction is the natural remediation. Prices use terminal fields without total-return adjustment, and the cost model, though deliberately conservative, is a stylized proxy for realistic execution. Finally, the study is confined to one market and one signal family; whether the regime effect replicates in other equity markets, and whether slower fundamental signals behave differently, are questions for future work.

VII. CONCLUSION

On a broad, recent panel of US equities we find that daily return-level prediction is a null result for every model considered, an LSTM included, and that a cross-sectional reformulation reveals statistically significant predictability that is concentrated in high-volatility regimes and stronger for nonlinear models. That predictability is nonetheless a one-day effect that survives neither transaction costs at daily frequency nor the transition to a weekly horizon. The signal is therefore real but not economically exploitable at the frequencies studied. We view the value of the work as methodological as much as empirical: with careful leakage controls, chronological evaluation and an explicit cost accounting, the honest answer is a precisely characterized negative result, which is more credible and more useful than an unaudited backtest that appears to succeed.

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